5058 Homework3

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1. Conditioning reduces entropy

(a)

The marginal probabilities $P(X)$ are found by summing the joint probabilities over all y:

$$\begin{aligned}
P(X=1) &= \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{4} = \frac{1}{2} \\
P(X=2) &= \frac{1}{16} + \frac{1}{8} + \frac{1}{16} + 0 = \frac{1}{4} \\
P(X=3) &= \frac{1}{32} + \frac{1}{32} + \frac{1}{16} + 0 = \frac{1}{8} \\
P(X=4) &= \frac{1}{32} + \frac{1}{32} + \frac{1}{16} + 0 = \frac{1}{8}
\end{aligned}$$

Thus, $P(X) = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right\}$.

The marginal probabilities $P(Y)$ are found by summing over all x:

$$\begin{aligned}
P(Y=1) &= \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{1}{4} \\
P(Y=2) &= \frac{1}{16} + \frac{1}{8} + \frac{1}{32} + \frac{1}{32} = \frac{1}{4} \\
P(Y=3) &= \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4} \\
P(Y=4) &= \frac{1}{4} + 0 + 0 + 0 = \frac{1}{4}
\end{aligned}$$

Thus, $P(Y) = \left\{\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right\}$.

(b)

The joint entropy $H(X,Y)$ is calculated using the non-zero probabilities in the table: one $\frac{1}{4}$, two $\frac{1}{8}$'s, six $\frac{1}{16}$'s, and four $\frac{1}{32}$'s.

$$\begin{aligned}
H(X,Y) &= -\sum_{x,y} p(x,y)\log_2 p(x,y) \\
&= -\left(1 \times \frac{1}{4}\log_2\frac{1}{4} + 2 \times \frac{1}{8}\log_2\frac{1}{8} + 6 \times \frac{1}{16}\log_2\frac{1}{16} + 4 \times \frac{1}{32}\log_2\frac{1}{32}\right) \\
&= -\left(\frac{1}{4}(-2) + \frac{1}{4}(-3) + \frac{3}{8}(-4) + \frac{1}{8}(-5)\right) \\
&= \frac{2}{4} + \frac{3}{4} + \frac{12}{8} + \frac{5}{8} \\
&= \frac{27}{8} = 3.375 \text{ bits}
\end{aligned}$$

(c)

The marginal entropies are:

$$\begin{aligned}
H(X) &= -\sum_{x} P(x)\log_2 P(x) \\
&= -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + 2 \times \frac{1}{8}\log_2\frac{1}{8}\right) \\
&= \frac{1}{2} + \frac{2}{4} + \frac{6}{8} = \frac{7}{4} = 1.75 \text{ bits}
\end{aligned}$$

$$\begin{aligned}
H(Y) &= -\sum_{y} P(y)\log_2 P(y) \\
&= -4 \times \left(\frac{1}{4}\log_2\frac{1}{4}\right) = 2 \text{ bits}
\end{aligned}$$

(d)

The conditional probabilities are calculated using $P(X|Y=y) = \frac{P(X,Y=y)}{P(Y=y)} = 4 \times P(X, Y=y)$.

- For $y = 1$: $P(X|Y=1) = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\}$.

$$H(X|Y=1) = 1.75 \text{ bits}$$

- For $y = 2$: $P(X|Y = 2) = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\}$.

$$H(X|Y = 2) = 1.75 \text{ bits}$$

- For $y = 3$: $P(X|Y = 3) = \{\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\}$.

$$H(X|Y = 3) = \log_2 4 = 2 \text{ bits}$$

- For $y = 4$: $P(X|Y = 4) = \{1, 0, 0, 0\}$.

$$H(X|Y = 4) = 0 \text{ bits}$$

(e)

Given $H(X) = 1.75$ bits:

(i) $H(X|Y = y) < H(X)$ occurs for $y = 4$ (since $0 < 1.75$).

(ii) $H(X|Y = y) > H(X)$ occurs for $y = 3$ (since $2 > 1.75$).

(f)

The conditional entropies are:

$$\begin{aligned} H(X|Y) &= \sum_y P(Y = y)H(X|Y = y) \\ &= \frac{1}{4}(1.75) + \frac{1}{4}(1.75) + \frac{1}{4}(2) + \frac{1}{4}(0) \\ &= \frac{5.5}{4} = \frac{11}{8} = 1.375 \text{ bits} \end{aligned}$$

$$H(Y|X) = H(X,Y) - H(X) = 3.375 - 1.75 = 1.625 \text{ bits}$$

(g)

The mutual information is:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 1.75 + 2 - 3.375 = 0.375 \text{ bits}$$

Verifying the identities:

$$H(X) - H(X|Y) = 1.75 - 1.375 = 0.375$$
$$H(Y) - H(Y|X) = 2 - 1.625 = 0.375$$

Thus, $I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$ is verified.

2. Principle of maximum entropy

(a)

The principle of maximum entropy states that the probability distribution which best represents the current state of knowledge is the one with the largest entropy, subject to known constraints.

(i) No other information is given
The only constraint is the normalization condition: $\sum_{i=1}^{6} p_i = 1$. To maximize $H = -\sum p_i \log p_i$ under this constraint, we use the uniform distribution:

$$p_i = \frac{1}{6}, \quad \text{for } i = 1, 2, \ldots, 6$$

(ii) Average is known to be $25/6$
Constraints: $\sum p_i = 1$ and $\sum_{i=1}^{6} i p_i = \frac{25}{6}$. The distribution that maximizes entropy takes the form:

$$p_i = \frac{e^{-\lambda i}}{Z}, \quad \text{where } Z = \sum_{j=1}^{6} e^{-\lambda j}$$

The parameter λ is determined by the constraint:

$$\frac{\sum_{i=1}^{6} i e^{-\lambda i}}{\sum_{j=1}^{6} e^{-\lambda j}} = \frac{25}{6}$$

Since $25/6 \approx 4.17$ is greater than the fair dice average (3.5), λ will be a negative value, shifting the probability mass towards larger faces.

(b)

To find the distribution $\{p_1, \ldots, p_n\}$ that maximizes the Shannon entropy

$$H = -\sum_{i=1}^{n} p_i \ln p_i$$

subject to

$$\sum_{i=1}^{n} p_i = 1 \quad \text{and} \quad \sum_{i=1}^{n} p_i \epsilon_i = \bar{\epsilon}$$

we define the Lagrangian:

$$L(p_1, \ldots, p_n, \lambda_0, \lambda_1) = -\sum_{i=1}^{n} p_i \ln p_i - \lambda_0 \left(\sum_{i=1}^{n} p_i - 1 \right) - \lambda_1 \left(\sum_{i=1}^{n} p_i \epsilon_i - \bar{\epsilon} \right)$$

Taking the partial derivative with respect to p_i and setting it to zero:

$$\frac{\partial L}{\partial p_i} = -\ln p_i - 1 - \lambda_0 - \lambda_1 \epsilon_i = 0$$

$$\ln p_i = -(1 + \lambda_0) - \lambda_1 \epsilon_i$$

$$p_i = e^{-(1+\lambda_0)} e^{-\lambda_1 \epsilon_i}$$

Applying the normalization constraint $\sum p_i = 1$:

$$e^{-(1+\lambda_0)} \sum_{i=1}^{n} e^{-\lambda_1 \epsilon_i} = 1 \implies e^{-(1+\lambda_0)} = \frac{1}{\sum_{j=1}^{n} e^{-\lambda_1 \epsilon_j}}$$

Thus,

$$p_i = \frac{e^{-\lambda_1 \epsilon_i}}{\sum_{j=1}^{n} e^{-\lambda_1 \epsilon_j}}$$

In statistical mechanics, for a system at temperature T, the multiplier λ_1 is identified as $\frac{1}{kT}$. Substituting this, we obtain the Maxwell-Boltzmann distribution:

$$p_i = \frac{e^{-\frac{\epsilon_i}{kT}}}{\sum_{j=1}^{n} e^{-\frac{\epsilon_j}{kT}}}$$

This distribution maximizes the Shannon entropy under the given mean energy constraint.

3. Differential entropy

Given a continuous random variable X with density $f(x)$, the differential entropy is defined as:

$$H(X) = -\int f(x) \log f(x) \, \mathrm{d}x$$

(a)

For $X \sim U(0, a)$, the density is $f(x) = 1/a$ for $0 \le x \le a$.

$$H(X) = -\int_0^a \frac{1}{a} \log\left(\frac{1}{a}\right) \mathrm{d}x = \frac{\log a}{a} \int_0^a \mathrm{d}x = \log a$$

If $a < 1$, then $H(X) = \log a < 0$. Unlike discrete entropy, differential entropy can be **negative**. This occurs when the distribution is highly "peaked" or concentrated in a region smaller than unit length.

(b)

For a normal distribution $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$:

$$\begin{aligned}
H(X) &= E[-\log f(X)] \\
&= E\left[\frac{1}{2}\log\left(2\pi\sigma^2\right) + \frac{X^2}{2\sigma^2}\log e\right] \\
&= \frac{1}{2}\log\left(2\pi\sigma^2\right) + \frac{E[X^2]}{2\sigma^2}\log e
\end{aligned}$$

Since $E[X^2] = \sigma^2$:

$$H(X) = \frac{1}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2}\log e = \frac{1}{2}\log\left(2\pi e\sigma^2\right)$$

(c)

For an exponential distribution $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$:

$$\begin{aligned}
H(X) &= -\int_0^\infty \lambda e^{-\lambda x}(\ln\lambda - \lambda x)\,\mathrm{d}x \quad \text{(assuming base } e\text{)} \\
&= -\ln\lambda \int_0^\infty \lambda e^{-\lambda x}\,\mathrm{d}x + \lambda \int_0^\infty \lambda x e^{-\lambda x}\,\mathrm{d}x \\
&= -\ln\lambda + \lambda E[X]
\end{aligned}$$

Since $E[X] = 1/\lambda$ for an exponential distribution:

$$H(X) = 1 - \ln\lambda = \ln(e/\lambda)$$

(d)

For a Laplace distribution $f(x) = \frac{1}{2}\lambda e^{-\lambda|x|}$:

$$\begin{aligned} H(X) &= E[-\log f(X)] = E\left[\log(2/\lambda) + \lambda|X|\right] \\ &= \log(2/\lambda) + \lambda E[|X|] \end{aligned}$$

For the Laplace distribution, $E[|X|] = \int_{-\infty}^{\infty} |x| \frac{\lambda}{2} e^{-\lambda|x|}\, \mathrm{d}x = \int_0^{\infty} \lambda x e^{-\lambda x}\, \mathrm{d}x = 1/\lambda$. Thus,

$$H(X) = \log(2/\lambda) + 1 = \log(2e/\lambda)$$

4. Relative entropy

(a)

To show $f(a) = \log a - a + 1 \leq 0$ for $a > 0$: Taking the first derivative with respect to a:

$$f'(a) = \frac{1}{a} - 1$$

Setting $f'(a) = 0$ gives the critical point $a = 1$. The second derivative is:

$$f''(a) = -\frac{1}{a^2}$$

Since $f''(a) < 0$ for all $a > 0$, $a = 1$ is a global maximum. The maximum value is:

$$f(1) = \log(1) - 1 + 1 = 0$$

Therefore, $f(a) \leq 0$ for all $a > 0$, and the equality $f(a) = 0$ holds **if and only if** $a = 1$.

(b)

The relative entropy is $D(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$. We consider:

$$-D(p \parallel q) = \sum_x p(x) \log \frac{q(x)}{p(x)}$$

Using the result from (a), $\log a \le a - 1$ where we let $a = \frac{q(x)}{p(x)}$:

$$\begin{aligned} -D(p \parallel q) &\le \sum_x p(x) \left(\frac{q(x)}{p(x)} - 1 \right) \\ &= \sum_x q(x) - \sum_x p(x) \\ &= 1 - 1 = 0 \end{aligned}$$

Thus, $-D(p \parallel q) \le 0 \implies D(p \parallel q) \ge 0$. The equality holds iff $p(x) = q(x)$ for all x.

(c)

Consider a binary alphabet $X = \{0, 1\}$.

(i) p(0) = 0.3, q(0) = 0.7
Then $p(1) = 0.7, q(1) = 0.3$.

$$\begin{aligned} D(p \parallel q) &= 0.3 \log_2 \frac{0.3}{0.7} + 0.7 \log_2 \frac{0.7}{0.3} \\ &= (0.7 - 0.3) \log_2 \frac{0.7}{0.3} = 0.4 \log_2 \frac{7}{3} \approx 0.4889 \text{ bits} \end{aligned}$$

Similarly,

$$\begin{aligned} D(q \parallel p) &= 0.7 \log_2 \frac{0.7}{0.3} + 0.3 \log_2 \frac{0.3}{0.7} \\ &= 0.4 \log_2 \frac{7}{3} \approx 0.4889 \text{ bits} \end{aligned}$$

In this specific case, $D(p \parallel q) = D(q \parallel p)$.

(ii) When does $D(p \parallel q) = D(q \parallel p)$ happen?
For a binary source, let $p(0) = p$ and $q(0) = q$. The equality holds when:

$$p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}$$

Rearranging the terms:

$$(p + q) \log \frac{p}{q} + (2 - p - q) \log \frac{1 - p}{1 - q} = 0$$

This occurs in two cases:

1. $p = q$: The two distributions are identical (both sides are 0).
2. $p = 1 - q$: The distributions are symmetric (as seen in part i).